Chapter 13 Notes

Section 1

Notes

Note that, on page 170, the separation axioms are listed incorrectly. They should use the formula $\phi(y, y_1, ..., y_n)$, not $\phi(x, y_1, ..., y_n)$.

Exercise 1.1 (a)

We prove that K has property F_2' iff all the separation axioms are true over K.

(a) On the one hand, suppose that K has property F_2'. Now consider any set $x \in K$ and any formula $\phi(y, y_1, ..., y_n)$ (for some elements $y_1, ..., y_n$ of K). Then let A be the class of all y such that $\phi(y, y_1, ..., y_n)$ is true over K (A is defined by $\phi(y, y_1, ..., y_n)$ over K). By F_2', $A \cap x \in K$. But $A \cap x$ satisfies the formula in **Ax 2**. Thus, for each formula $\phi(y, y_1, ..., y_n)$ and $x \in K$, the separation axiom for that formula and element holds over K.

(b) On the other hand, suppose that all the separation axioms are true over K. Then consider any first-order definable subclass A of K and any element b of K. A must be definable over K by some formula $\phi(x)$. Then, by the separation axiom, there is some $z \in K$ such that, for all $x \in K$, $x \in z \equiv (x \in b \wedge \phi(x))$. This formula works as, if $x \in b \cap A$, then $x \in A$, and thus $x \in K$, meaning the formula would hold true. Of course, if the formula is true, then $x \in b \cap A$.

Exercise 1.1 (b)

We prove that K (where K is transitive) is a first-order universe iff all the axioms of ZF are true over K.

(a) On the one hand, suppose that K is a first-order universe. We show that all the ZF axioms are true over K.

Ax 1: Suppose that $(\forall z)(z \in x \equiv z \in y)$. Then, because K is transitive (by F_1), it follows that $x = y$.

Ax 2: We show that this holds by showing that F_2' holds (in K), and consequently **Ax 2** would hold by Exercise 1.1 (a). So suppose A is a first-order definable subclass of K and b is an element of K. Then, because K is first-order swelled, it follows that $A \cap b \in K$ (because $A \cap b$ is first-order definable over K, as it is defined by the formula $(y \in z \equiv (y \in x \wedge \phi(y)))$ where $\phi(y)$ is some formula that defines A over K).

Ax 3: This follows by F_3.

Ax 4: This follows by F_4.

Ax 5: This follows by F_5, as K is transitive. This is because, as K is transitive, the set y defined by

$$(\exists y)(\forall z)(z \in y \equiv (\exists w)(z \in w \wedge w \in x))$$

is precisely the union of x.

Ax 6: This follows by F_6. For, if $x \in K$, then $\mathcal{P}(x) \cap K \in K$. But $\mathcal{P}(x) \cap K$ is precisely the y defined by the formula

$$(\exists y)(\forall z)(z \in y \equiv z \subseteq x)$$

Ax 7: This follows by F_7.

Ax 8: This follows by F_8.

Ax 9: This follows by F_9.

(b) On the other hand, suppose that all the ZF axioms are true over K. We show that K is a first order universe.

F_1: This holds by assumption.

F_2: Consider any $x \in K$ and some first-order definable (over K) subset A of x. By Lemma 1.1, condition F_2' holds, which means that $A \cap x = A$ is in K. Thus K is first-order swelled.

F_3: This follows by **Ax 3** (because K is transitive).

F_4: This follows by **Ax 4** (again because K is transitive).

F_5: This follows by **Ax 5** (again because K is transitive).

F_6: This follows by **Ax 6**.

F_7: This follows by **Ax 7**.

F_8: If F is a function that is first-order definable over K, then it is defined by some formula $\phi(x, y)$. Then, for every $w \in K$, by **Ax 8** there is a $v \in K$ such that

$$(\forall y)(y \in v \equiv (\exists x)(x \in w) \wedge \phi(x, y))$$

which is just saying $F''(w) \in K$.

F_9: This follows by **Ax 9** (because K is transitive).

Exercise 1.2

It is not clear what is being asked here. Are the ZF axioms being discussed the first-order ones, or the ones introduced at the beginning of the book? If they are the former, clearly we cannot be using Theorems from chapters 7 and 8, where the first-order axioms were not yet introduced. For the problem, I am therefore assuming we are speaking of the intial ZF axioms. Well, suppose we add the axiom of well foundedness. By Theorem 3.5 of Chapter 8, it follows that $V = R_\Omega$, which is a Zermelo-Fraenkel Universe by Theorem 4.3 of Chapter 7. Thus these new axioms are consistent by assumption.

This means that the axiom of well foundedness cannot be disproved in ZF for the following reason: For suppose it could be. Then adding the axiom of well foundedness to the ZF axioms would introduce a contradiction, contrary to the result of this exercise.